

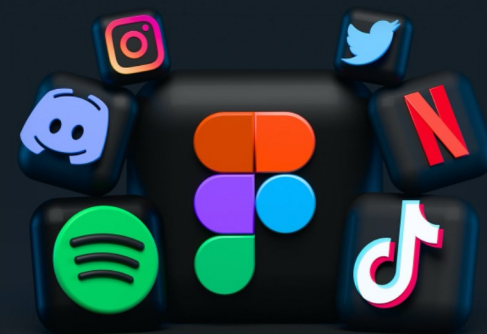
Studying radicalization & polarization on social media

Petter Törnberg



This talk

- Dominant polarization research
- Two limitations of existing research
- Examples of NLP-based research



A detailed historical painting of a 18th-century coffeehouse interior, featuring ornate wood paneling, a large chandelier, and several tables with patrons. The scene is overlaid with modern digital elements: numerous smartphones are being held by the people, and several red heart-shaped social media notification icons with numbers (11, 1, 1, 1, 1, 1, 1, 1) are floating in the air. The text is overlaid on the right side of the image.

A new public sphere?

Internet was to bring a new democratic renaissance - democratic deliberation and a new public sphere (Habermas 1962).

Remove gatekeepers, lower costs of participation, meeting as equals (Castells 2012, Benkler 2007)



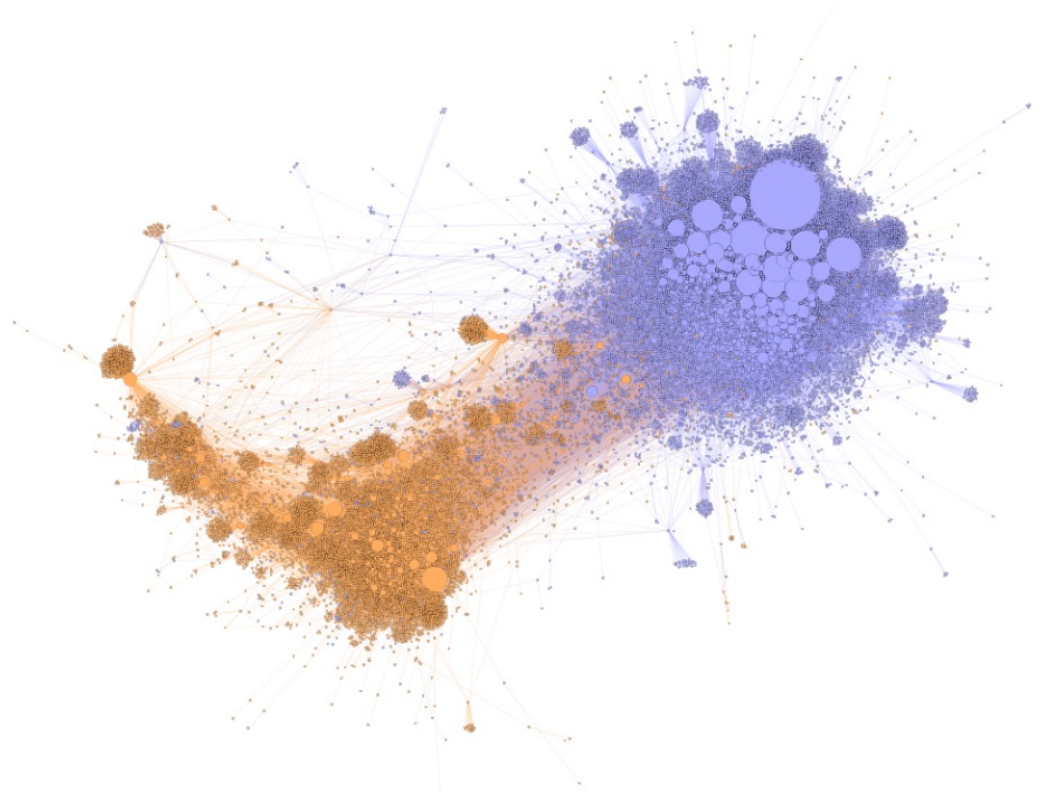
Democratic crisis

Rising polarization,
toxicity, and
misinformation

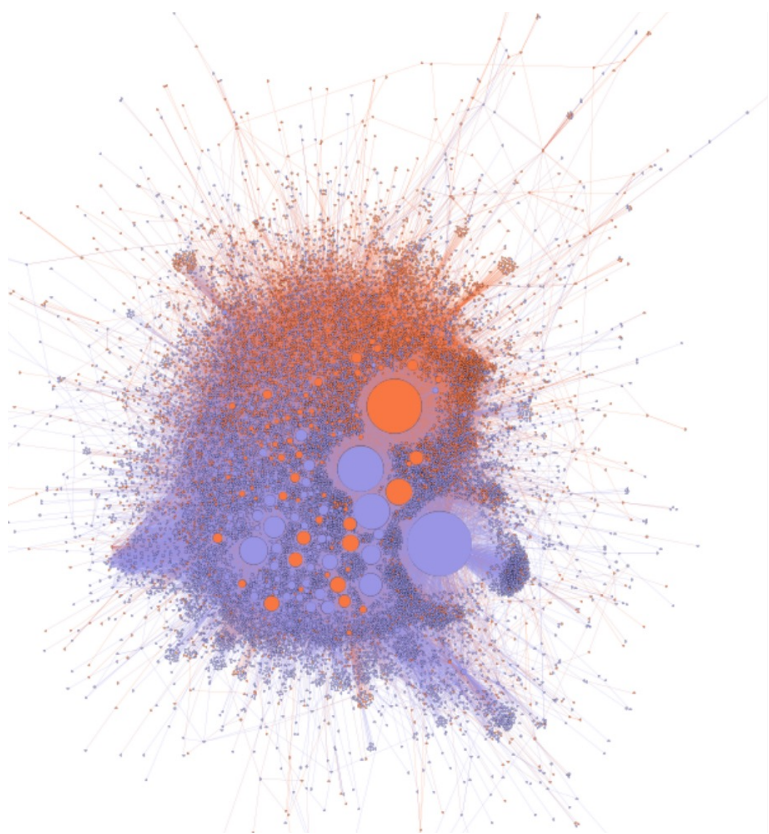
Echo chambers?

- Cass Sunstein (1999): social media cause us to segregate into homogeneous groups with likeminded others
- Isolation leads one-sided deliberation which leads to extremism and radicalization





Limitation 1: Conflating network cohesion with social cohesion



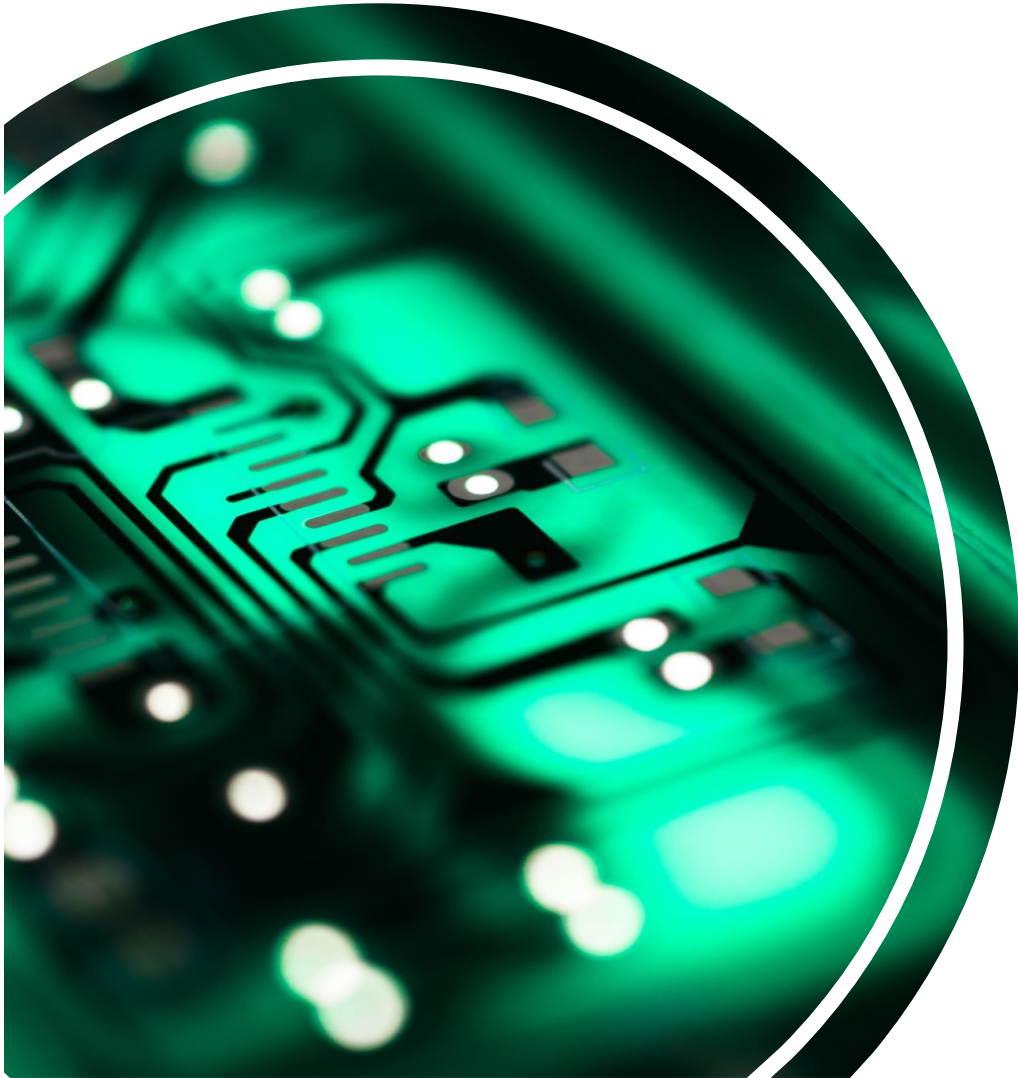
Limitation 1: Ignoring meaning

1. Single case has resulted in conflicting findings on whether echo chambers exist
2. Equates network homophily and polarization - but studies have shown that interaction with opponents can *increase* polarization (Bail et al 2018)

We need to go beyond the *structure* of interaction, to also looking at its *contents*.

From Social Network Analysis to Natural Language Processing

- Decades of digital research focused on the structure of networks – likes, shares, and followings
- Now, large-scale text analysis is becoming more and more feasible.



NATURAL LANGUAGE PROCESSING

- New field at the intersection of computer science, artificial intelligence, and linguistics
- One of the most important method of Computational Social Science
- Often doesn't fit into the classic quant-qual divide
- NLP can be used to 'predict', but can also be used as part of a quantitative or qualitative approach
- Predict, classify, cluster
- In-depth and large scale



Examples of NLP methods

- Sentiment analysis: Identify the feelings expressed in a given text.
- Topic modeling: Unsupervised method to identify themes from textual data.
- Machine Translation: Translate from one language to another
- Named Entity Recognition (NER): Identify all the people and objects in a text.
- Machine Learning: Make predictions, automatically annotate data.

NLP often involves a combination of several techniques to carry out some specific task.



Large Language
models are
taking over!

Prompts operationalize concepts

1. Define concept (clear codebook)
2. Design prompt (explicit instructions + examples)
3. Run annotation
4. Validate on labeled subset
5. Iterate prompt
6. Report:
 - prompt
 - model
 - validation metrics

Dis/Advantages with LLMs



1. Powerful, versatile, no training data required
2. Very easy to use using API: simple as getting the weather!
3. Bridges quant and qual by enabling computational interpretation
4. Works across languages and contexts!



1. More expensive and slower than supervised models.
2. Accuracy depends on task
3. Black-boxes
4. Reproducibility



The 8 commandments of LLM text annotation

1. Choose an appropriate model
2. Follow a structured coding process
3. Develop a Prompt Codebook
4. Validate your model
5. Use structured prompts
6. Consider ethical and legal implications
7. Prompt stability analysis
8. Consider that your data may be in the training data.

Limitation 2: Online phenomena in a vacuum

- Challenging to link online phenomena to specific social and political and geographical contexts.
- Most search for a hashtag. We end up studying phenomena as existing in a kind of digital vacuum – as unconnected from social and political context
- As if there was such a thing as an “online phenomenon.”





Single-platform, single-country cases

- Hashtag samples.
- Non-representative, unclear link to social political context.

Consequences:

- Difficult to link to existing theory.
- Treating “social media” as monolithic.
- Studying digital phenomena in a vacuum

Törnberg, P., & Söderström, O. (2025). Comparative platform urbanism: Cities in a world of platforms. *Digital Geography and Society*, 8, 100119.

Example studies



Comparative data: Twitter Parliamentary database

- All Twitter posts from all parliamentarians in 26 countries, 2017-2022:
32M tweets
- Manually collated
- Streaming API – full data.
- Python, PostgreSQL.
- Connected to ParlGov, CHES, V-Dem, etc. for comparative analysis



Van Vliet, L., Törnberg, P., & Uitermark, J. (2020). The Twitter parliamentary database: Analyzing Twitter politics across 26 countries. *PLoS one*, 15(9), e0237073.

Study 1: Is politics becoming more uncivil?

- Political incivility leads to lower trust and legitimacy, weaker deliberation, polarization, and higher tolerance for political violence.
- General sense that politics incivility is rising – *but is it? And if so, what drives the rise?*

Törnberg, P., & Chueri, J. (2025). Elite Political Discourse has Become More Toxic in Western Countries. arXiv preprint arXiv:2503.22411.

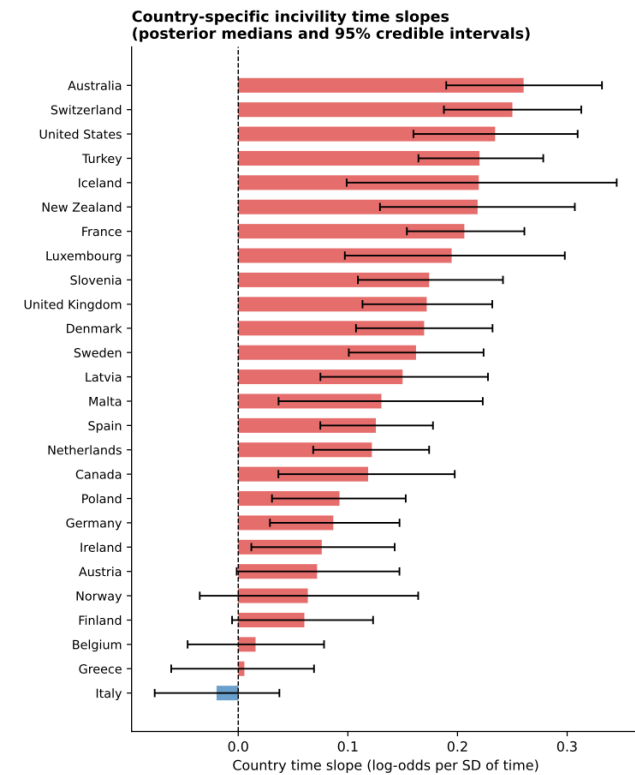
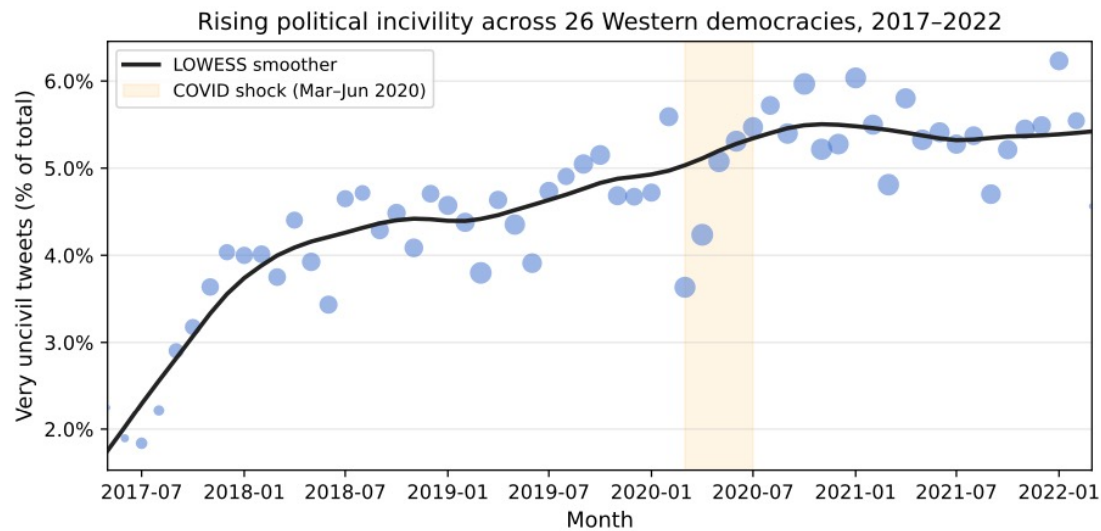


Method

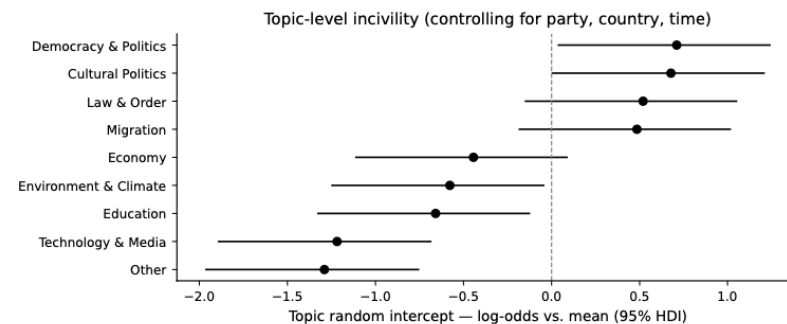
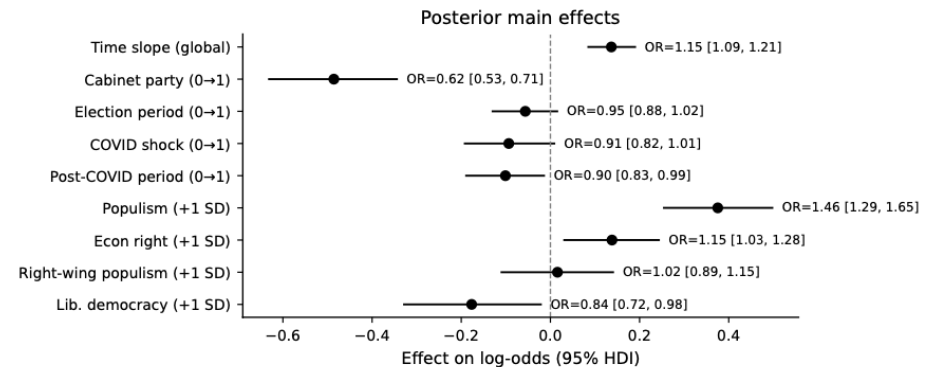
- LLM-based coding of incivility: 3 levels.
- Validated against human coding.
- **Dataset:** 18M tweets from MPs in 26 countries over 5 years



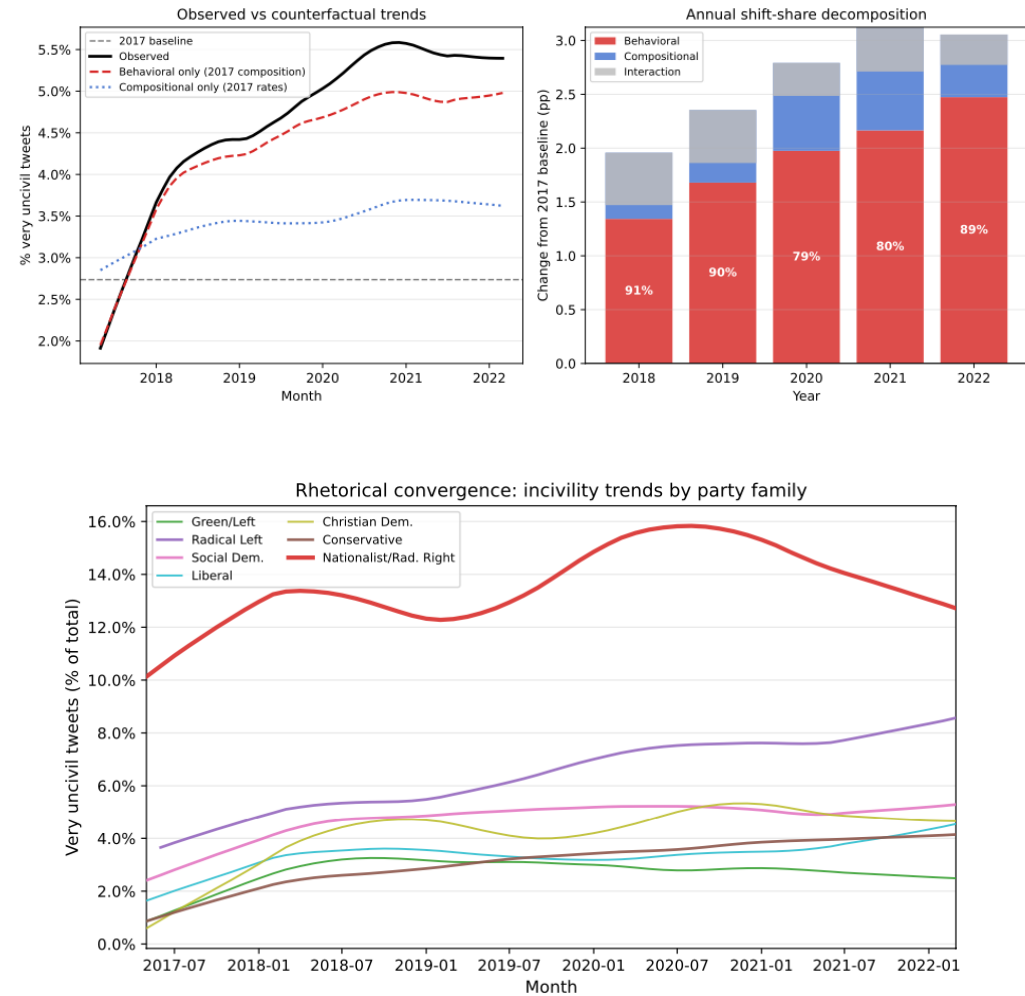
Incivility is rising



- Populist and right-wing politics are strong predictors for incivility
- Cultural issues and democracy are associated to higher incivility.



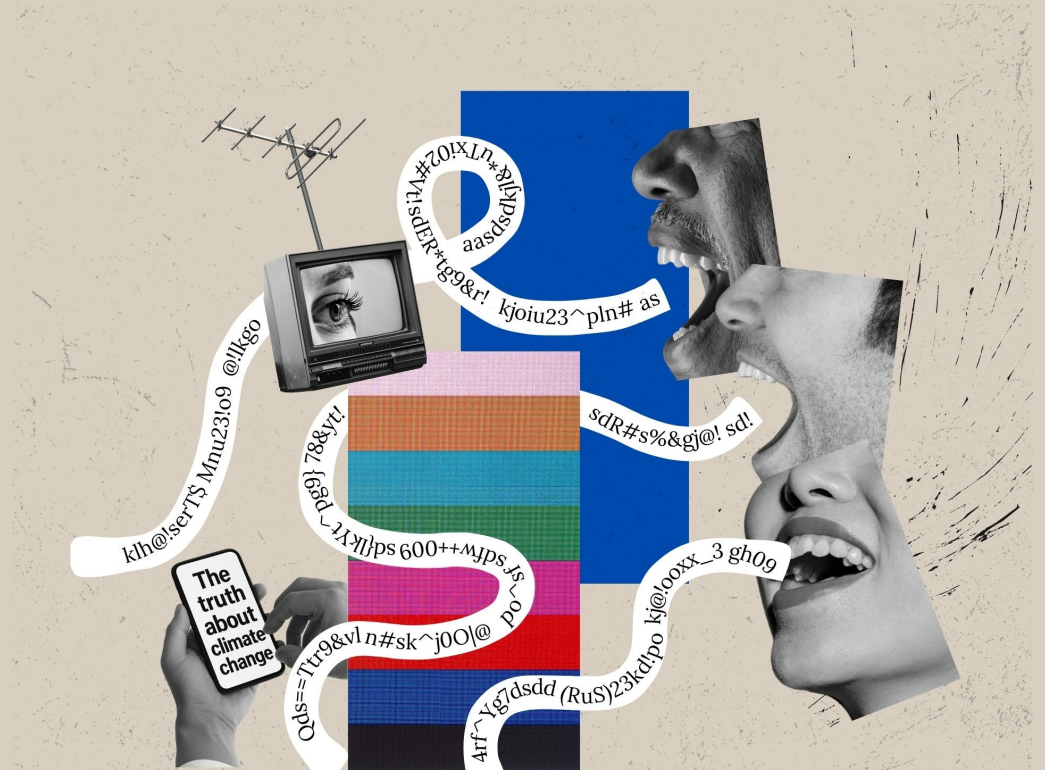
- The rise in incivility is behavioral: not explained by a shift in composition
- The rise consists mostly of other parties catching up with the radical right.



Conclusions

- Radical right sticks out.
- Toxicity is rising sharply: 100% increase.
- Findings are compatible with changing structural conditions.
- Shifting incentives driven by social media: calculated approach to attention in an increasingly crowded information ecosystem.

Study 2: Political elites and misinformation



The background of the slide is a dark, textured illustration. It features several laptops and tablets, some of which are engulfed in bright orange and yellow flames. The overall theme is 'digital wildfire', representing the rapid spread of misinformation through digital media.

Misinformation as a 'digital wildfire'?

- Misinformation research exploded after the 2016 U.S. election and Brexit.
- Misinformation was seen a social media phenomenon (Humprecht et al., 2019; Lazer et al., 2018).
- Falsehoods spread "farther, faster, deeper" than true information (Vosoughi et al., 2018), like "wild-fire over social networks" (Törnberg, 2018).
- Information condition: lack of access to veracity.

Less common than you think?

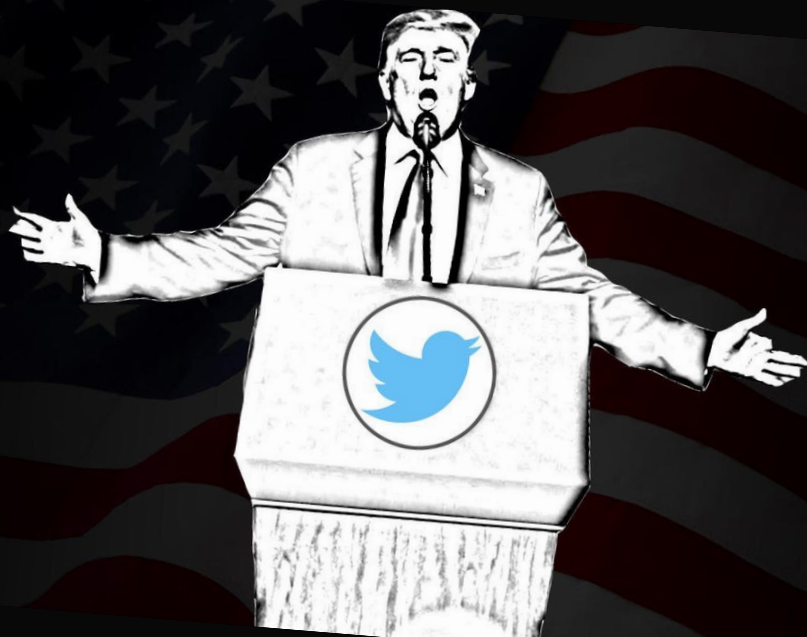


- But more recent studies suggest that most people in fact do not consume or share misinformation (Guess et al 2020). “Moral panic”.
- Yet, misinformation is widespread among specific electoral groups:
 - A majority of Republicans believing Trump won the 2020 election.
- Problems with our conception of misinformation?

Misinformation in a
social and political
vacuum



Our argument: Misinformation is a political phenomenon



- Misinformation is not a media phenomenon, but a strategy of political elites.
- Misinformation is part and parcel of particular forms of politics.
- Spreading misinformation can be strategic:
 - Control public attention
 - Agenda-setting
 - Distract voters and media
 - Shape public opinion
 - Energize and mobilize base by reinforcing existing beliefs, fears or resentments
 - Cast political opponents in a negative light

New approach?

We need to study misinformation as
situated in political and social contexts:

Which parties spread misinformation?

Again, we need a comparative approach



Method

- **Data:** parliamentary database
- Identify when parliamentarians share misinformation:
 - Extract all URLs shared by parliamentarians
 - Compile databases of misinformation (e.g., MediaBiasFactCheck)
 - These provide levels of misinformation
 - Manually validated.
- **Party factuality score:** aggregation of degree of factuality on each party.
- **Model:** Beta-regression hierarchical model.
Controls for party internal cohesion, party participation in the government, and degree of party personalization



Hypotheses

- **H1:** Parties with higher populist ideology are more likely to spread misinformation on Twitter.
- **H2:** Parties with higher scores on the left-right ideological scale are more likely to spread misinformation on Twitter.
- **H3:** Populist parties on the right side of the political spectrum are more likely to spread misinformation on Twitter than left-wing populist parties.

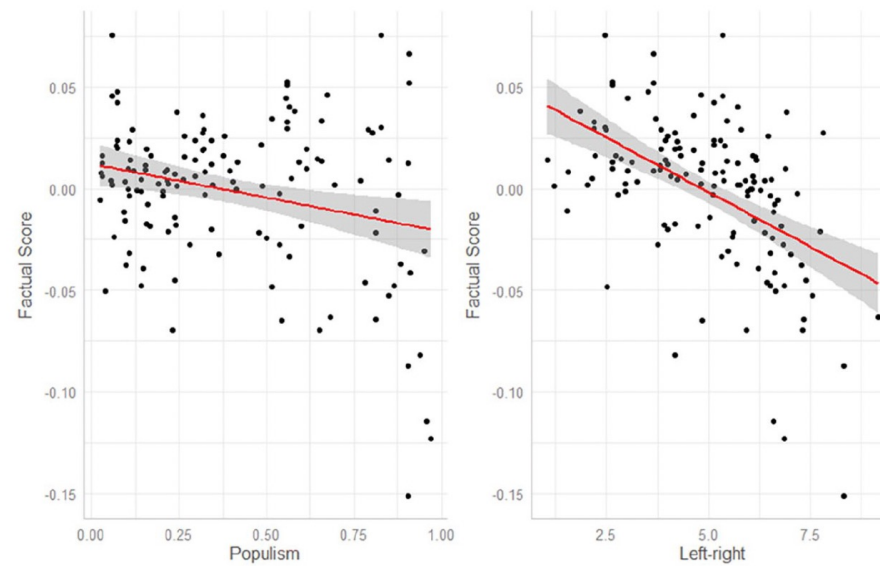
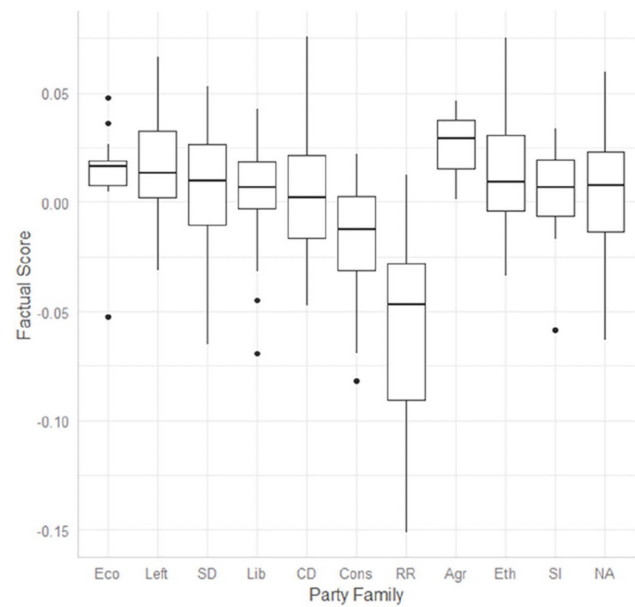


Figure 1. Association between factuality score adjusted by country mean, parties' left-right ideology (right), and populism (middle), and government participation (left) with 95% confidence intervals.

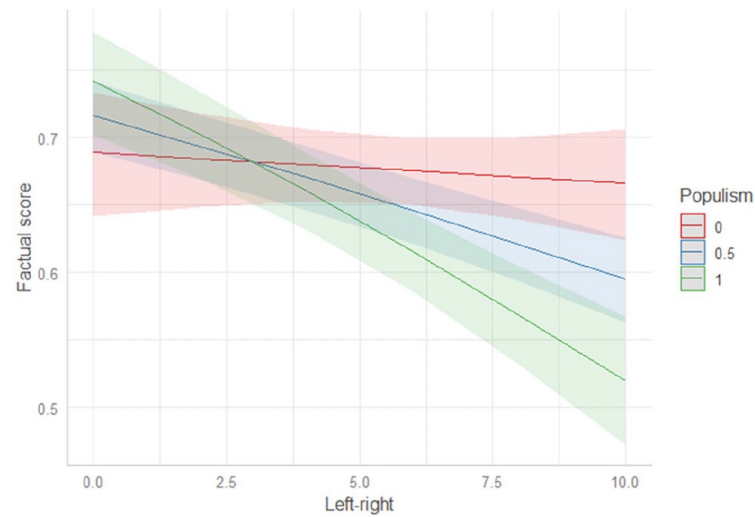
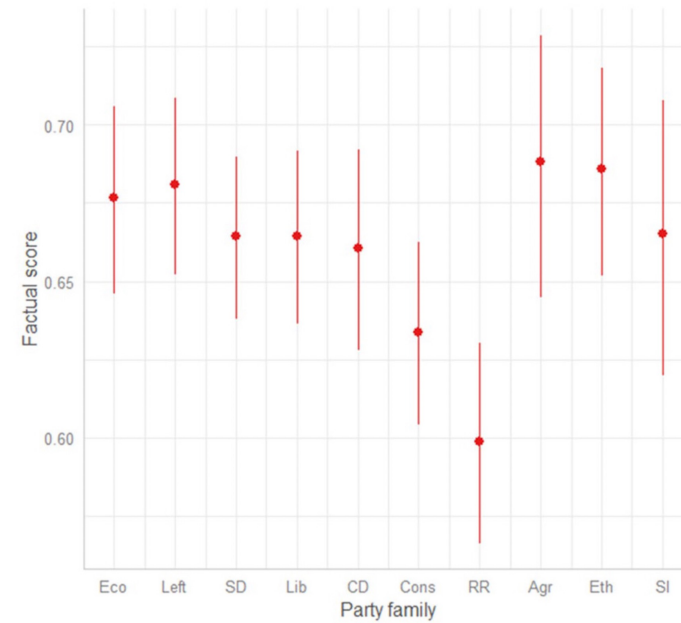


Figure 3. Predicted values of factuality score, by party position into the left-right scale and populist ideology with 95% confidence intervals.



Finding:

- Government/opposition is not significant.
- Populism is not in itself associated with misinformation.
- Only when it is combined with right-wing ideology

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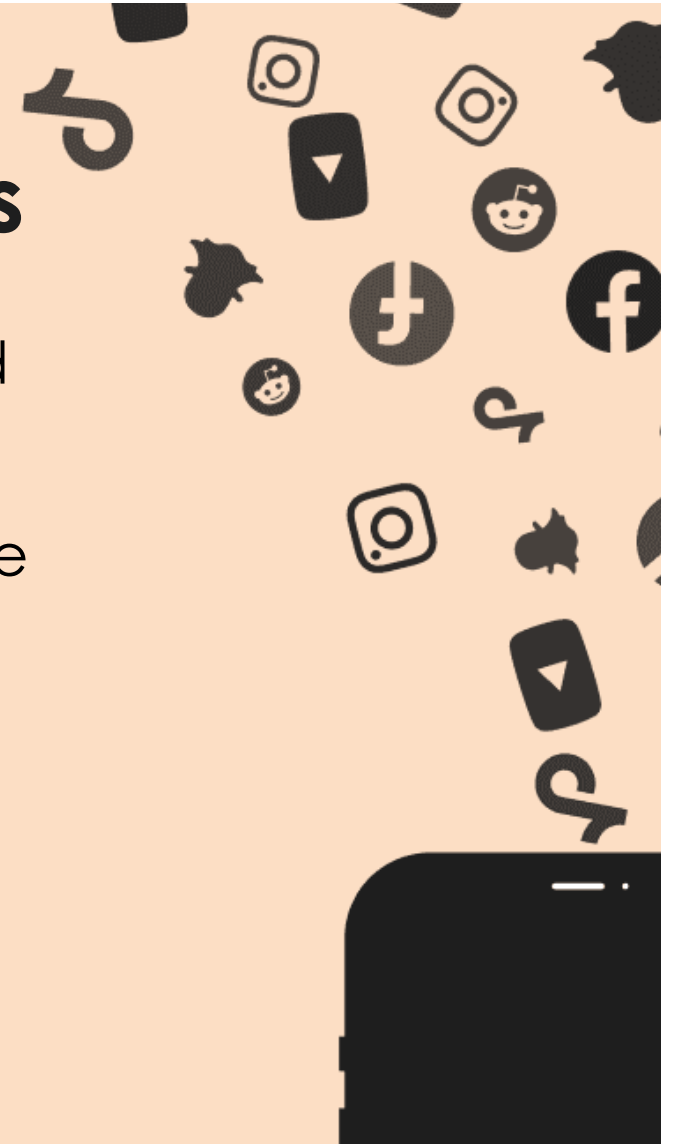
Conclusions study 2

- Neither left-right ideological position nor populist ideology alone explain misinformation: only combined.
- The problem is not populism but radical-right populism.
- Is misinformation a social media phenomenon – or *populist radical right phenomenon*?



Drawing together the studies

- Our methods shape how we understand digital phenomena
- Radical right drive misinformation and incivility - but incivility is also strong on the left
- PRRP are challenging democratic institutions, using misinformation as a strategy
- Polarization? Risk for false equivalence



The politics of attention

So are these *social media phenomena* – or *populist radical right phenomena*?

- Yes!

Social media is not “causing” polarization or misinformation: it shapes the political economy of attention.
This transforms *politics at large*.
(McLuhan and Postman)

Social media and the populist moment are inextricable.



Studying social media as if it was part of society

- Social media phenomena are also social and political phenomena – and must be studied as such.
- And conversely, social media must be understood as part of longer economic, political and social trends.





Next stop: Workshop!